BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Captures user/environment data.

Q2. Communicates information or causes action.

Q3. Output device controlling physical movement/action.

Q4. Six.

You Do pages · Independent questions

Q1. Sensor measures temperature [2], CPU compares threshold [2], actuator controls fan/heater [2].

Q2. Barcode cheap but line-of-sight [3], RFID can scan tags without direct optical sight but cost/interference [3].

Q3. Touch or keypad as input [2], audio navigation/physical buttons [2], readable display [2], spoken confirmation/printed ticket [2].

Exit Ticket · Model answer

False. It both displays and senses touch input.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

LESSON 103 · Lesson 103 · Input and Output: Selecting Devices for Tasks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an input device do?

Q2. What does an output device do?

Q3. What is an actuator?

Q4. A 2×3 display grid: how many cells?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

New learning

Understand — An input device captures data or a user action; an output device communicates information or changes physical conditions.

Apply — A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

Evaluate — Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Worked example

Recommend input and output for automated greenhouse temperature control.

1. Explain the choice — Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

We Do · Practise with guidance

A library kiosk needs to identify books and display return confirmation. Select suitable devices.

Explain the choice

Input barcode reader or RFID reader; output display and optional printer.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]

Question 3

Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

Exit Ticket · Answer before checking

A touchscreen can only be an output device.